

The background features a complex, abstract design. It consists of numerous thin, dark, wavy lines that sweep across the frame, creating a sense of motion and depth. Interspersed among these lines are various numbers (0-9) in a light gray font, some of which are slightly blurred or faded, giving the impression of data points or a digital landscape. The overall color palette is monochromatic, with shades of gray and white, accented by a small orange square on the left side of the title bar.

4. Data Exploration

Introduction

Objectives of Data Exploration

- Understanding data
- Data preparation
- Data mining tasks
- Interpreting data mining results

Introduction

Roadmap for data exploration

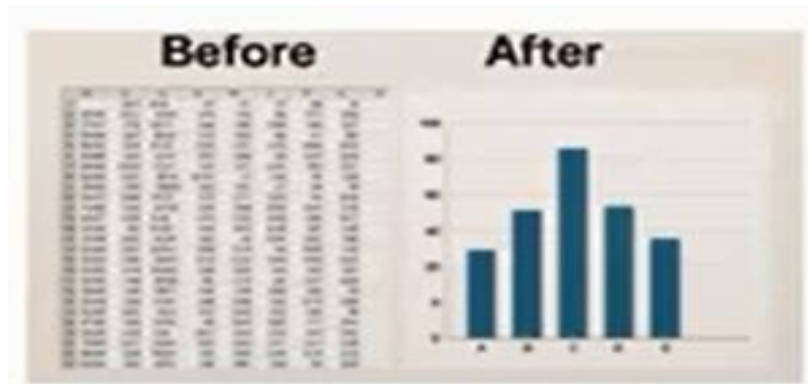
- Organize the data set
- Find the central point for each attribute
- Understand the spread of the attributes
- Visualize the distribution of each attributes
- Pivot the data
- Watch out for outliers
- Understanding the relationship between attributes
- Visualize the relationship between attributes
- Visualization high dimensional data sets

Data Visualization

Data visualization: is the practice of translating information into a visual context, such as a map or graph, to make data easier for the human brain to understand and pull insights from.

Why Visualize data?

- **Data is every where:** in newspapers, streets, schools – data surrounds us daily in graphs, tables, and images
- **Simplifying Complexity:** good visualization turn complex information into simple, powerful message anyone can understand.



Data Visualization

. Why Data Visualization Matters?

- In today flood information, **strong visualize help message stand out**, **connect emotionally** and **inspire action**.
- **Spotting Trends**: It's easy to see if sales are climbing or dipping over time.
- **Identifying Outliers**: Data points that don't fit the pattern (anomalies) stand out immediately.
- **Speed**: Decision-makers can grasp the "big picture" in seconds rather than hours of reading reports.

Master the Art of Visualize Data Analysis

- **Reduce information overload:** the first step in visual analysis is to filter out the noise and focus on what's essential.
- **Choose the right graph:** each type of graph – bar, line, pie, scatter – tells a different kind of story based on data.
- **Analyze and display relationships:** visuals make it easy to spot trends, patterns, and connections that raw numbers can hide.
- **Empower decision making:** strong data visualization doesn't just show information, it helps people make smarter, faster decisions.

How to Analyze One, Two Variables, or More?

- **Univariate Analysis**

- **Focus:** one variable
- **Goal:** describe, summarize, and find patterns
- **Visualization**



- Continuous: Histogram  , Box plot 

- Categorical: Bar chart  , Pie chart 

- **Bivariate Analysis**

- **Focus:** two variables
- **Goal:** explore relationships
- **Visualization**

- Continuous + Continuous: Scatter plot 

- Categorical: Table plot 

How to Analyze One, Two Variables, or More?

- **Multivariate Analysis**
 - **Focus:** 3+ variables
 - **Goal:** regression, clustering, and factor analysis
 - **Visualization**
 - 3D scatter plot

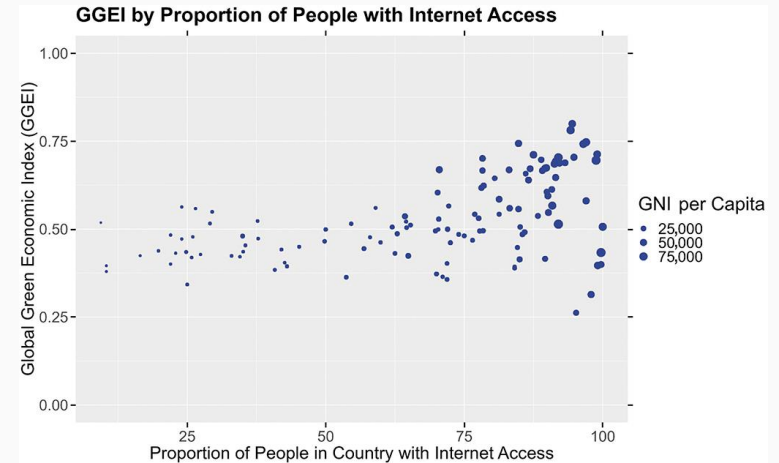


The Grammar of Graphics

Useful framework for describing and constructing visualizations

Elements of data visualizations:

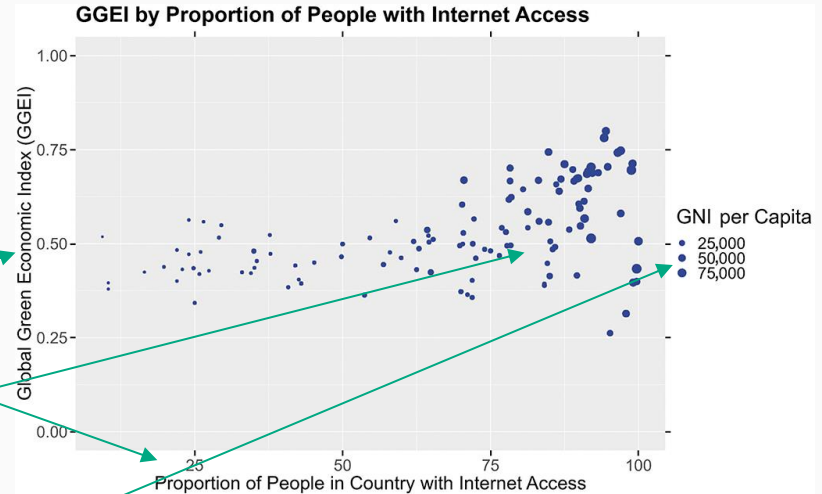
- source data set
- x-axis
- y-axis
- color
- geometry, or shape (inside the plotting region)



The Grammar of Graphics in Action

Elements of a data visualization are determined by the **source dataset** or by the data scientist's **preference**

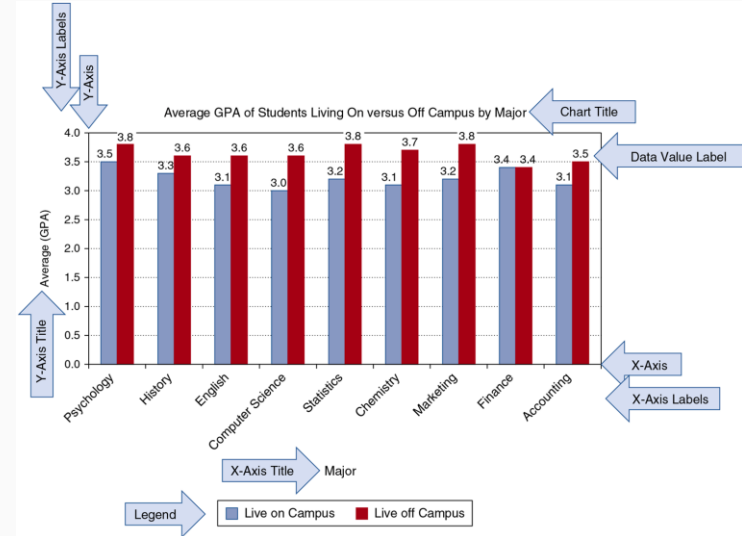
- **x-axis:** proportion of people with internet access
- **y-axis:** GGEI
- **color:** blue
- **geometry, or shape:** points
- **size (of points):** GNI Per Capita



Grammar of Graphics in General

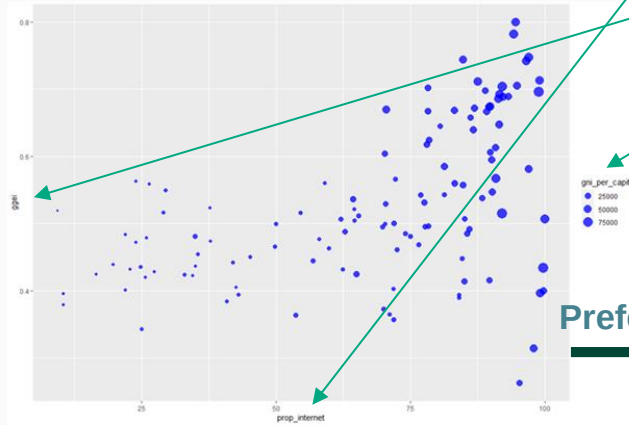
Common associations:

- x/y-axis: variable from our data set
- color: often a variable from our data set
- size: often a variable from our data set

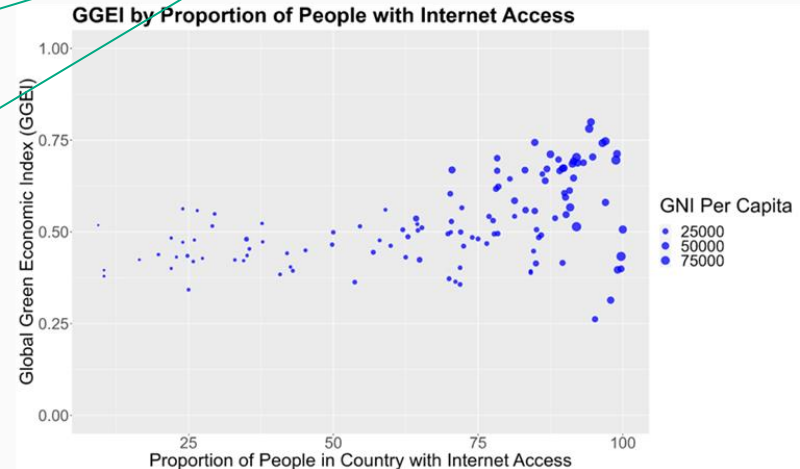


From Data to Visualization with the Grammar of Graphics

ID	iso_alpha	country	continent	income_g	life_expec	mean_yea	cemissior	under_5_n	prop_internet	whr_score	population	gni	gni_per_capita	ggei	pop_cat
1	ABW	Aruba	North Am	High income							106537				Less than 10 million
2	AFG	Afghanistan	Asia	Low income	61.9824	2.98507	0.363	58.1	18.4		40099462	73149074277	1824.190915		Greater than 10 million
3	AGO	Angola	Africa	Lower mic	61.6434	5.417391	1.03	72.3	36		34503774	1.89E+11	5465.617791		Greater than 10 million
4	ALB	Albania	Europe	Upper mic	76.4626	11.28646	2.01	8.52	72.2	53.6	2811666	39731962626	14131.11039	0.565677	Greater than 10 million
5	ARE	United Arab Emirates	Asia	High income	78.7104	12.69403	19.8	7.16	100	64.6	9365145	5.86E+11	62573.59181	0.506517	Greater than 10 million
6	ARG	Argentina	South Am	Upper mic	75.3899	11.14727	3.53	9.4	85.5	59	45608747	9.59E+11	20925.26814	0.48498	Greater than 10 million

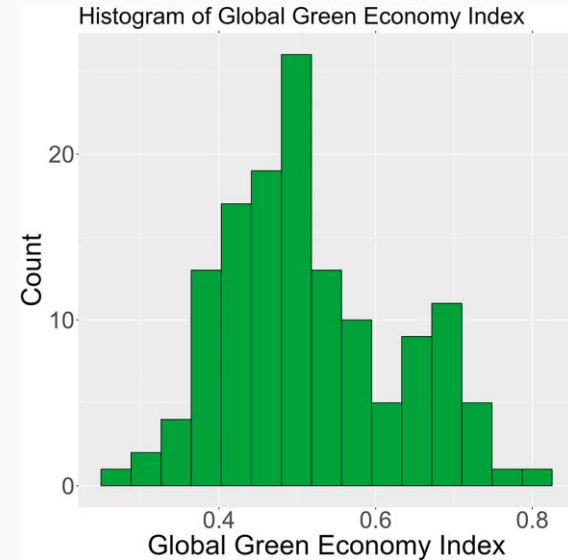


Preference



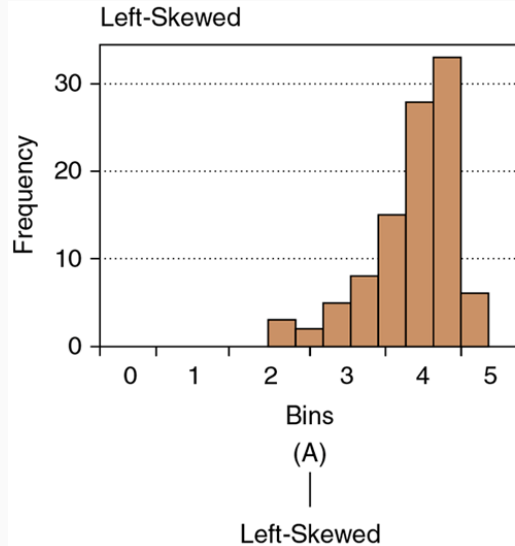
Histograms

- Display the distribution of continuous data
- Show frequency of data values within intervals
- Vary in shape
 - Symmetrical
 - Left-skewed
 - Right-skewed

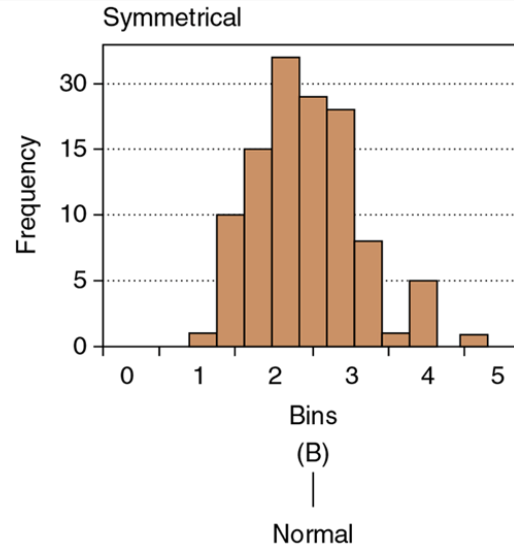


Histogram Symmetry and Skewness

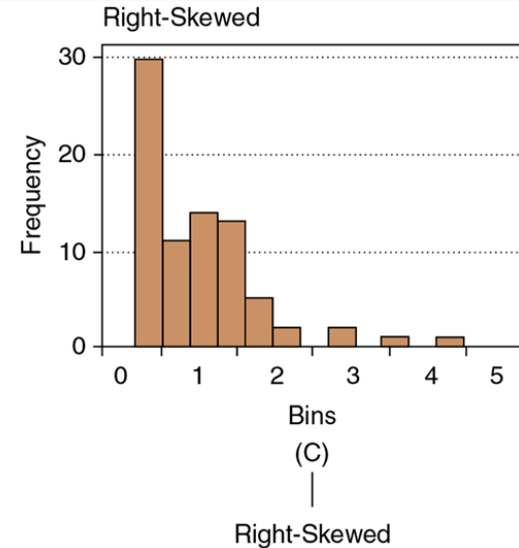
1



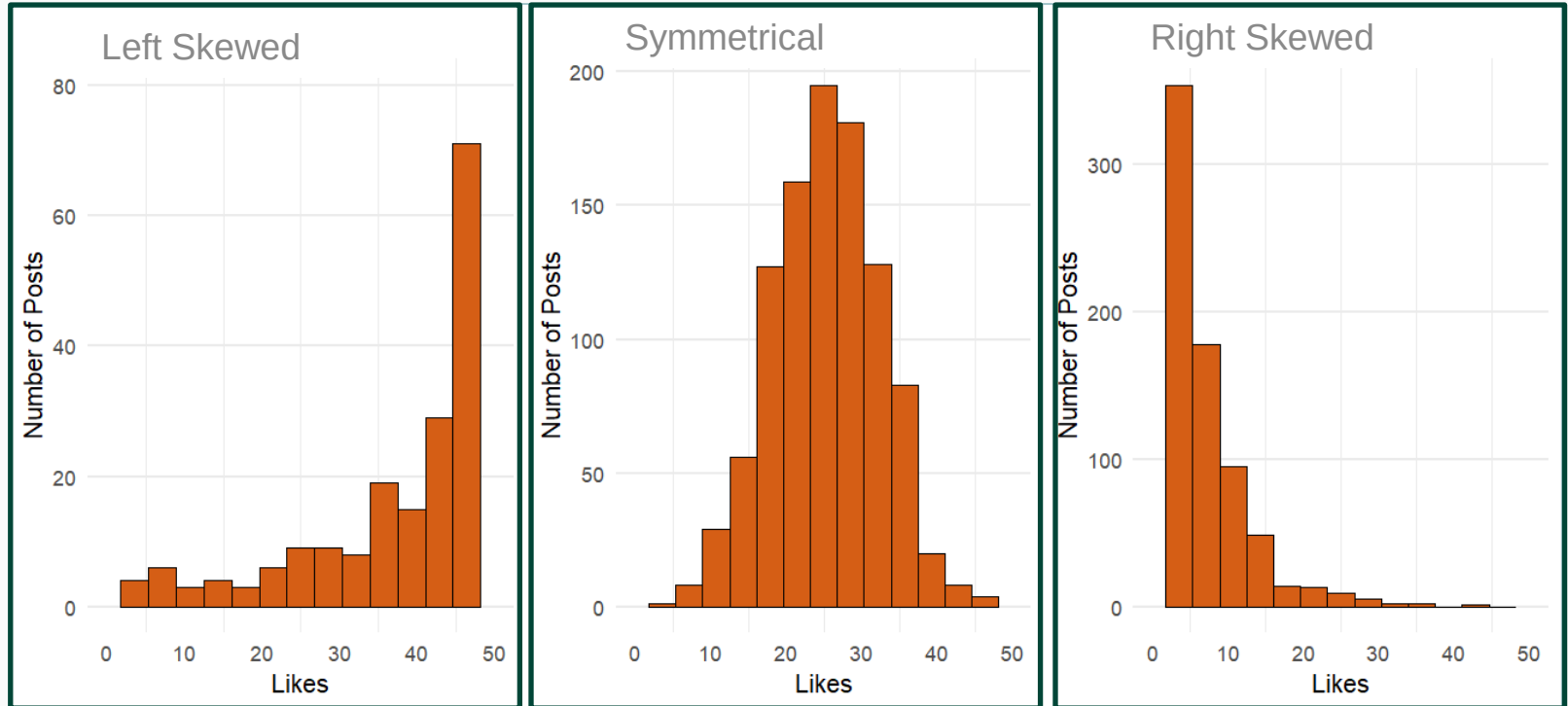
2



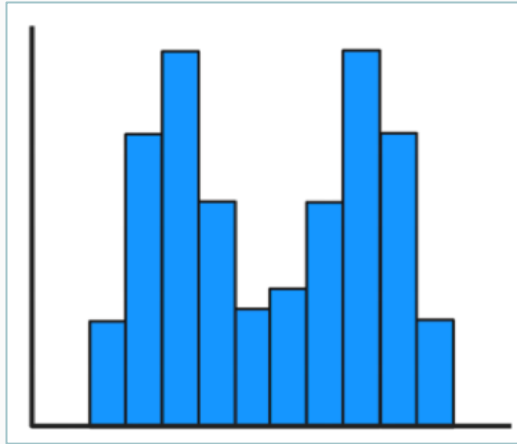
3



Applying Histograms to Social Media Analytics

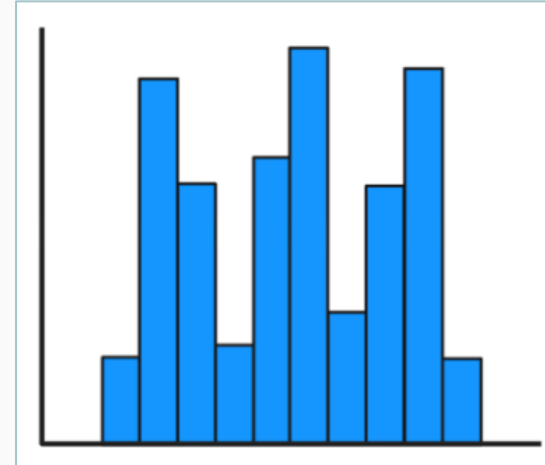


Bimodal and Multimodal Histograms



Bimodal Histogram

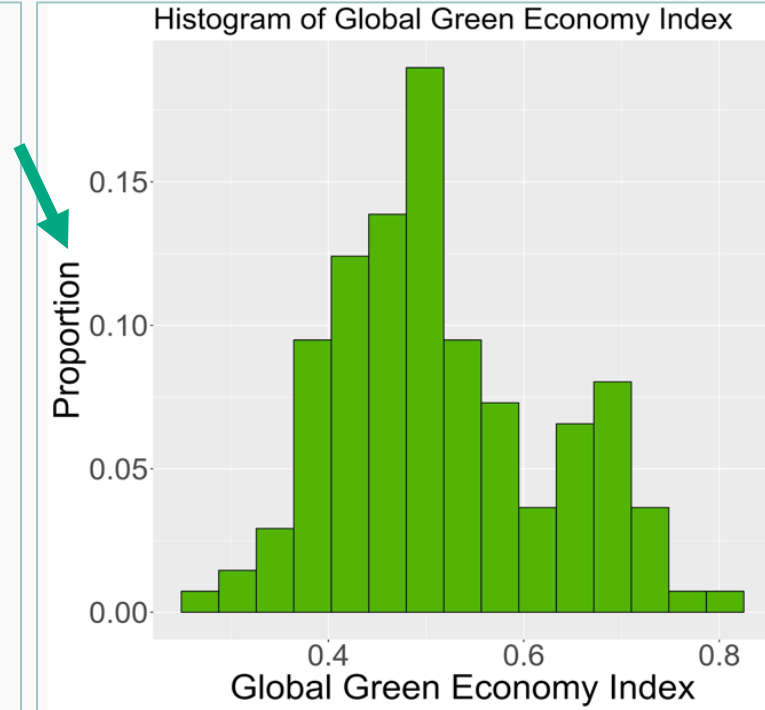
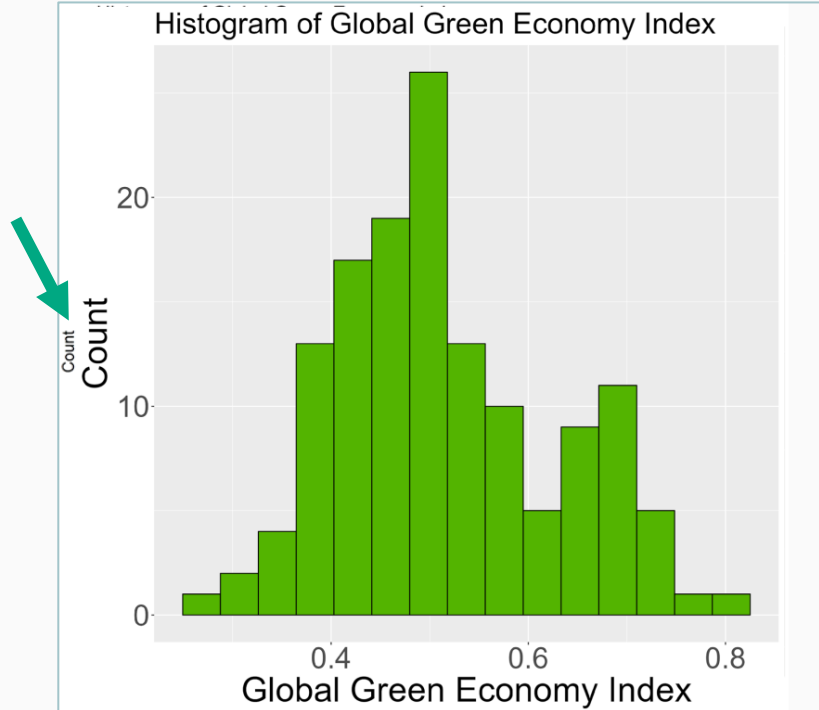
Two distinct peaks
Two different modes



Multimodal Histogram

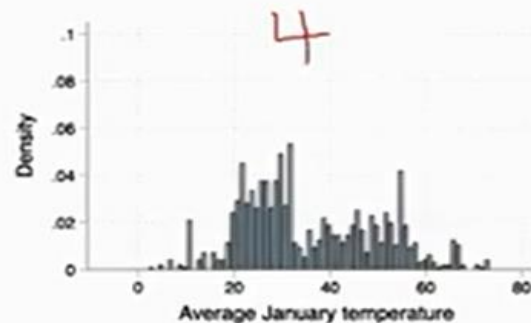
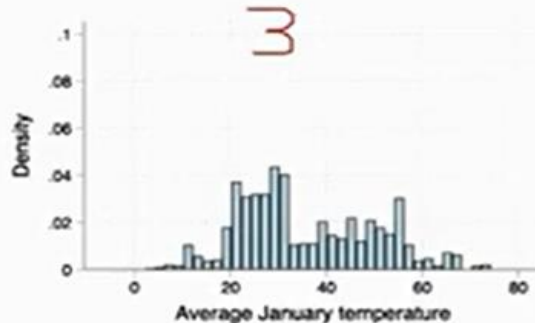
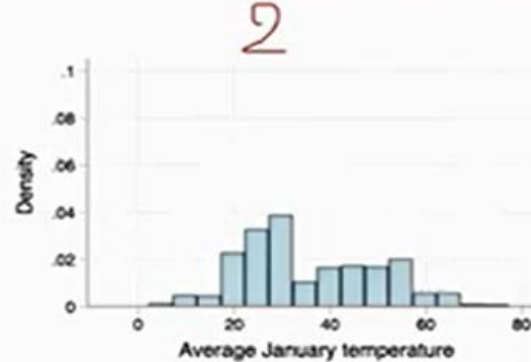
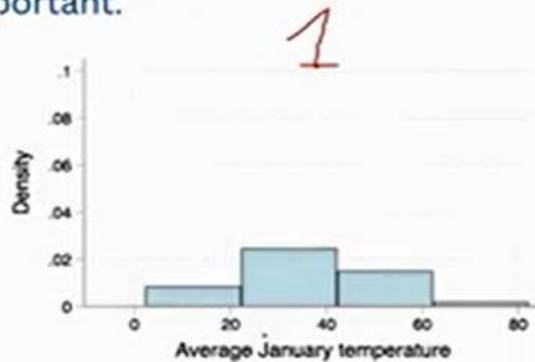
More than two peaks
Multiple modes

Exploring the Global Green Economy Index (GGEI) with Histograms



Bin Sizes of Histogram

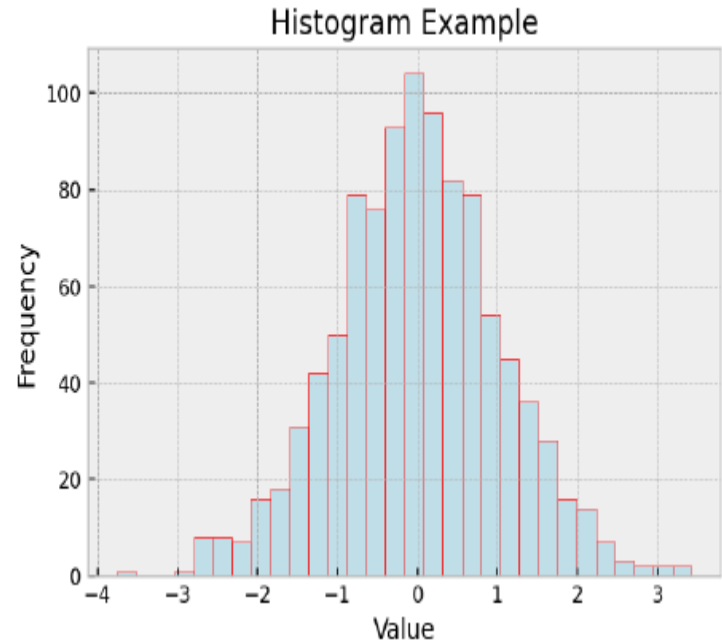
Bin sizes are crucial: Larger bins show less detail, while smaller bins can reveal more detail. Finding the right balance is important.



Histogram Code

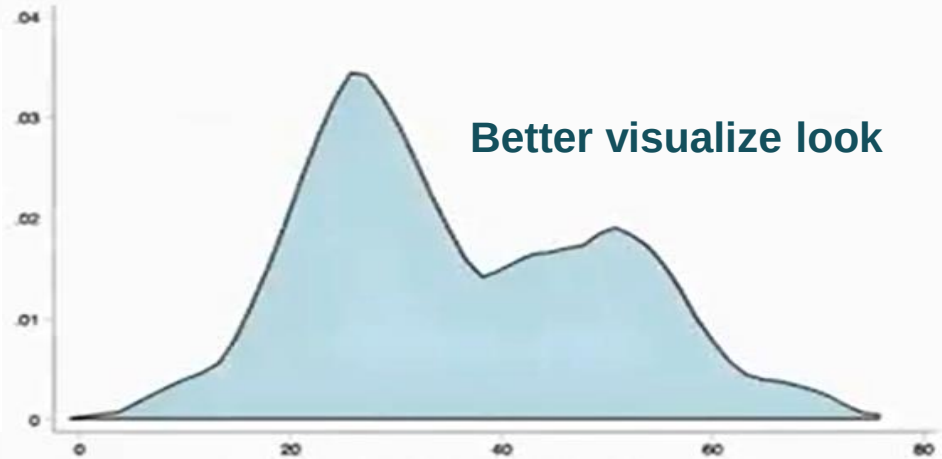
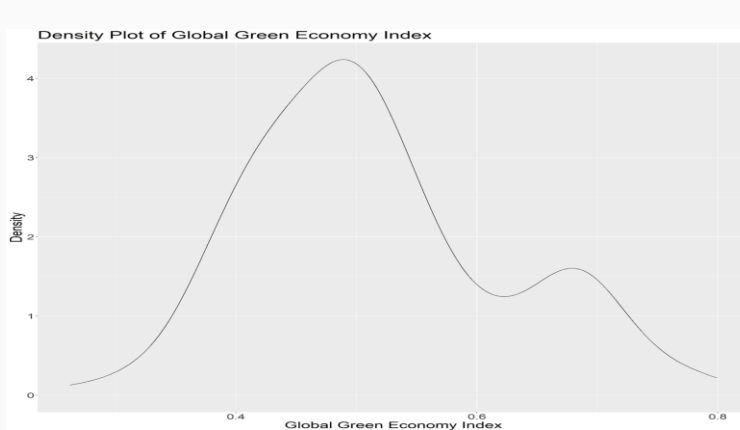
```
data = np.random.randn(1000) # 1000 random points from  
normal distribution
```

```
plt.hist(data, bins=30, color="lightblue", edgecolor="red",  
alpha=0.7)  
plt.title("Histogram Example")  
plt.xlabel("Value")  
plt.ylabel("Frequency")  
plt.show()
```



Density Plots

- Smoothed, continuous representation
- Identifies patterns and trends
- Enhances visual appeal

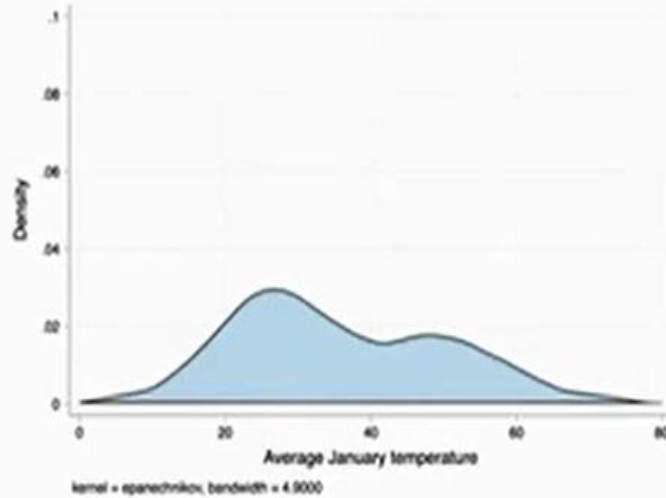


Density Plots

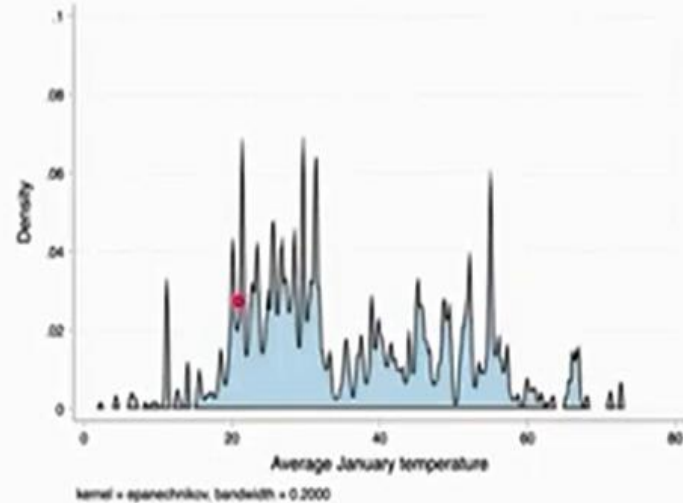
Parameters:

The shape of a density plot is influenced by bandwidth and kernel type. Higher bandwidths lead to flatter shapes, while lower bandwidths provide more detail. The default kernel types, like Epanechnikov or Gaussian, usually produce the best results.

Larger bandwidth = too little details

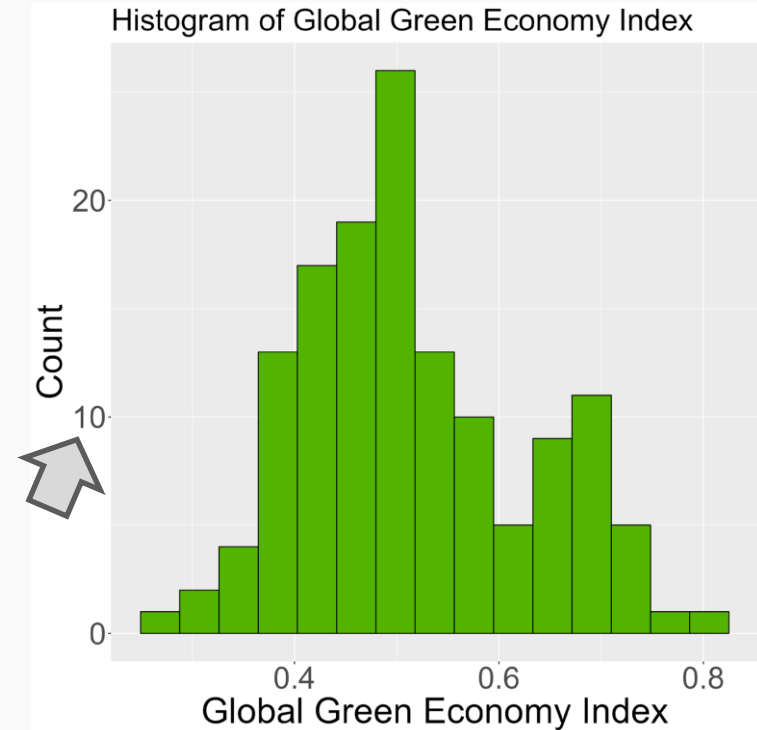
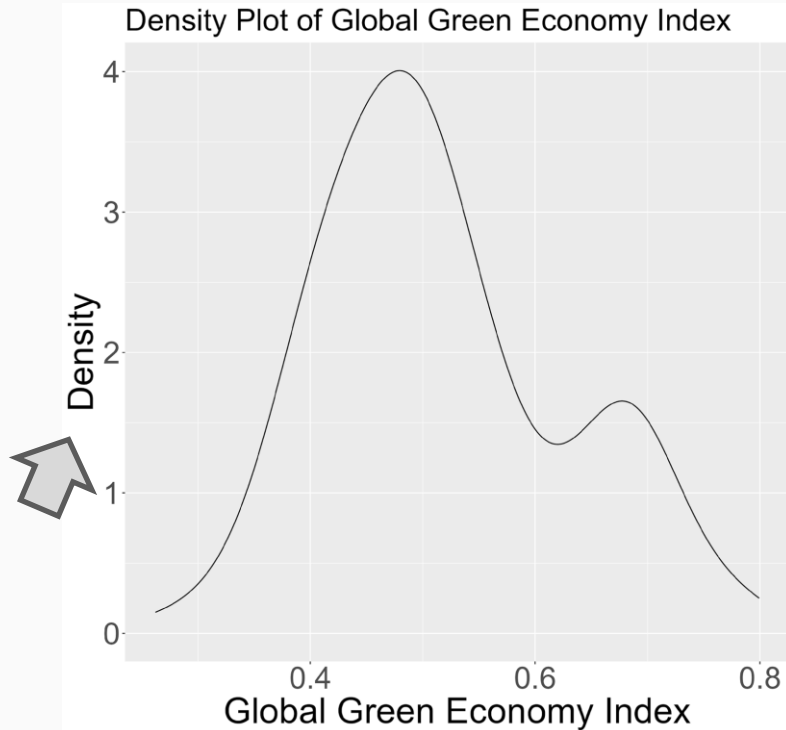


Small bandwidth = too much details

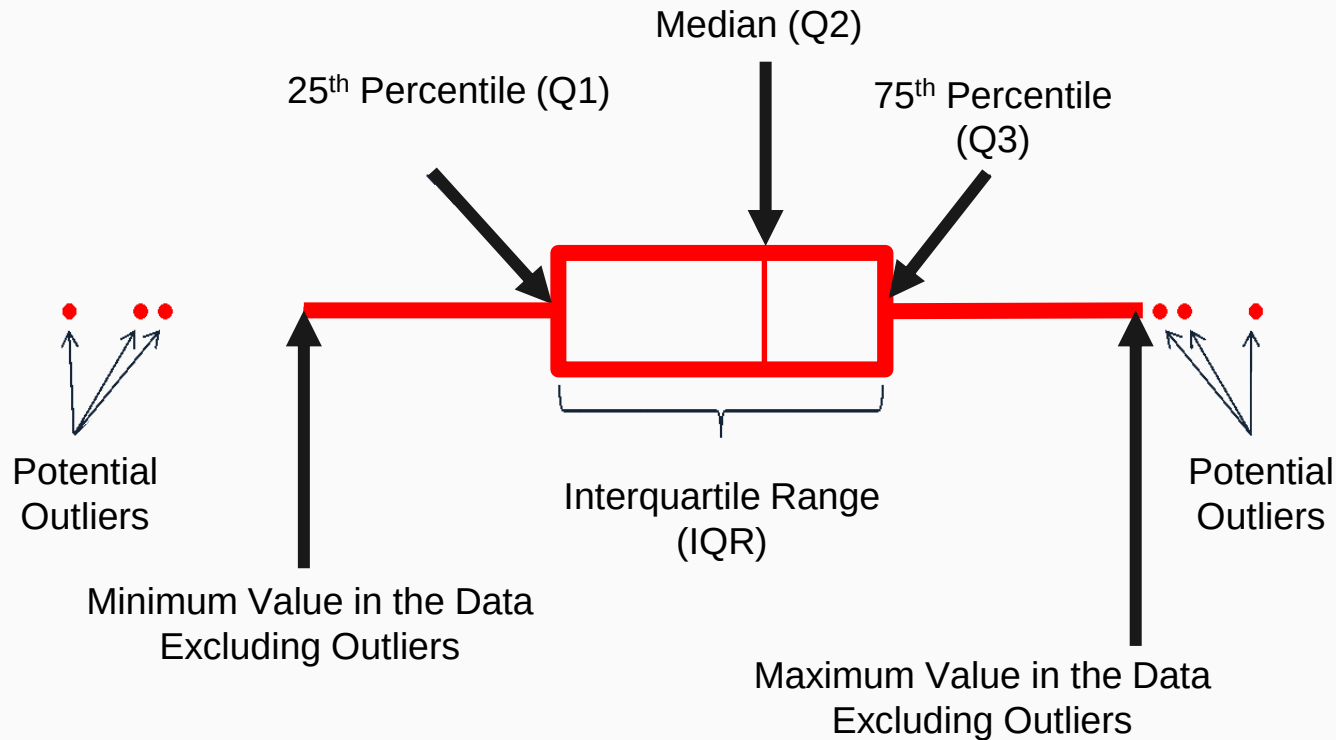


**Avoid changing the kernel
from its default.**

Comparing Histograms and Density Plots



Box Plots



Video here

Box Plot Example

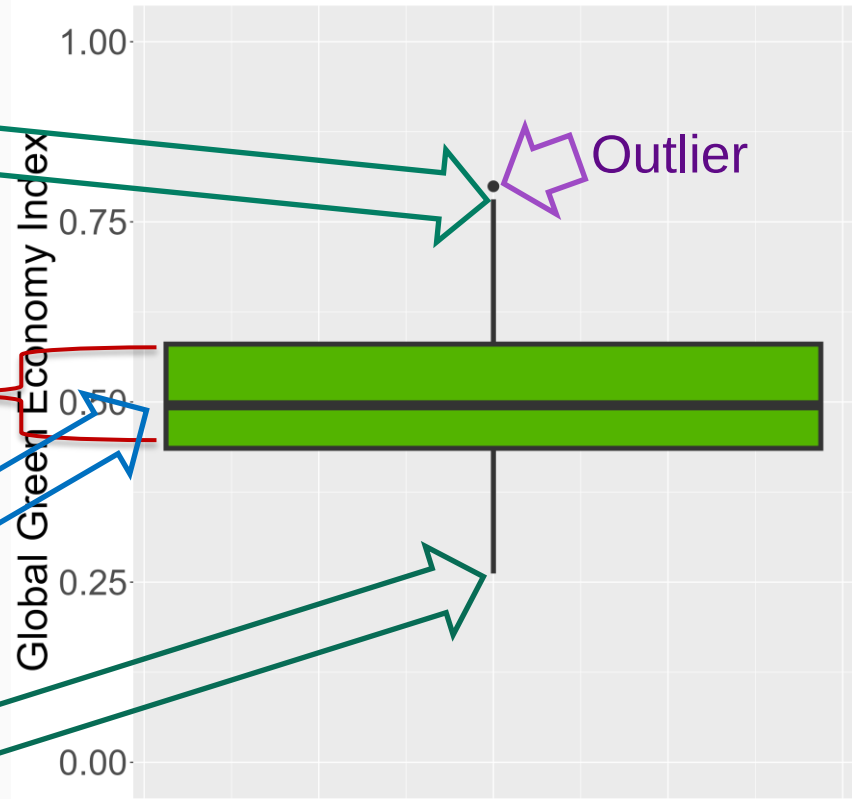
Maximum value whisker

Middle 50% of the Data

Median

Minimum value whisker

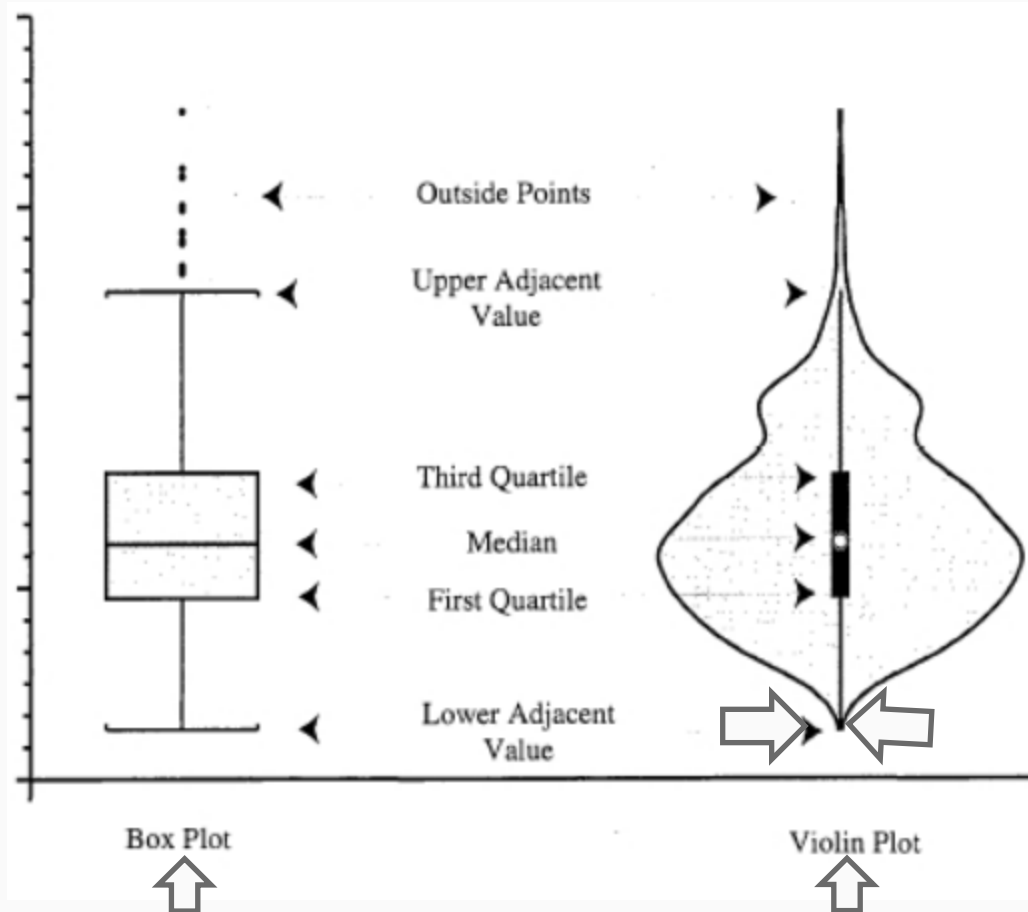
Boxplot of Global Green Economy Index



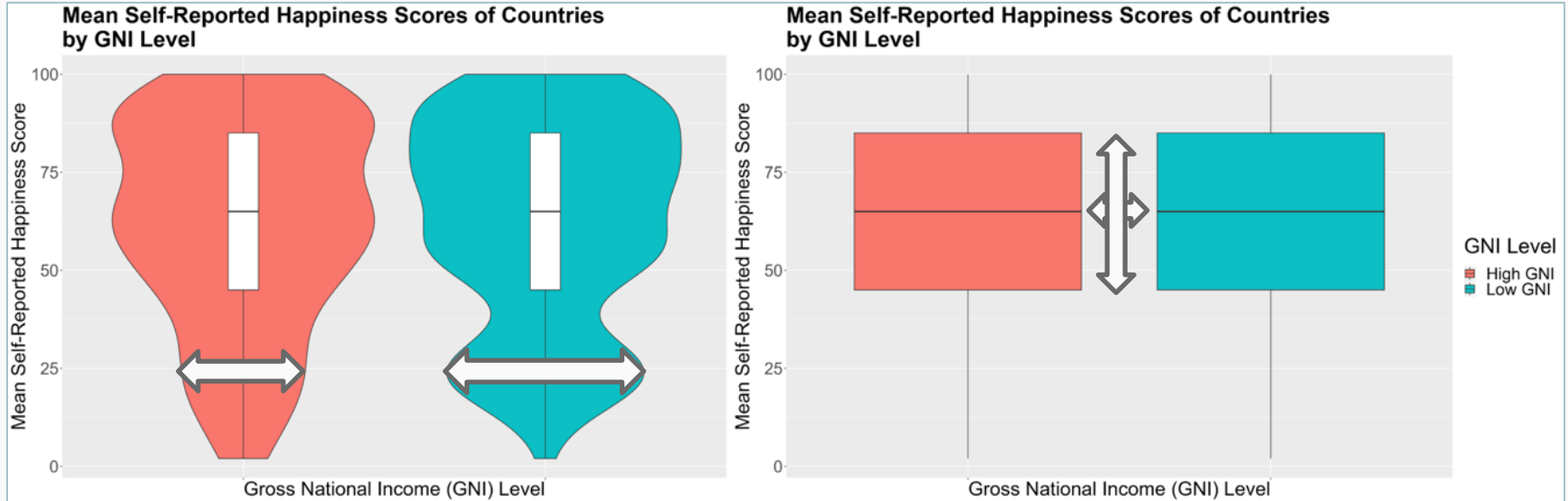
Violin Plots

- Combine features of box plots and density plots
- Provide a more detailed view of data distribution



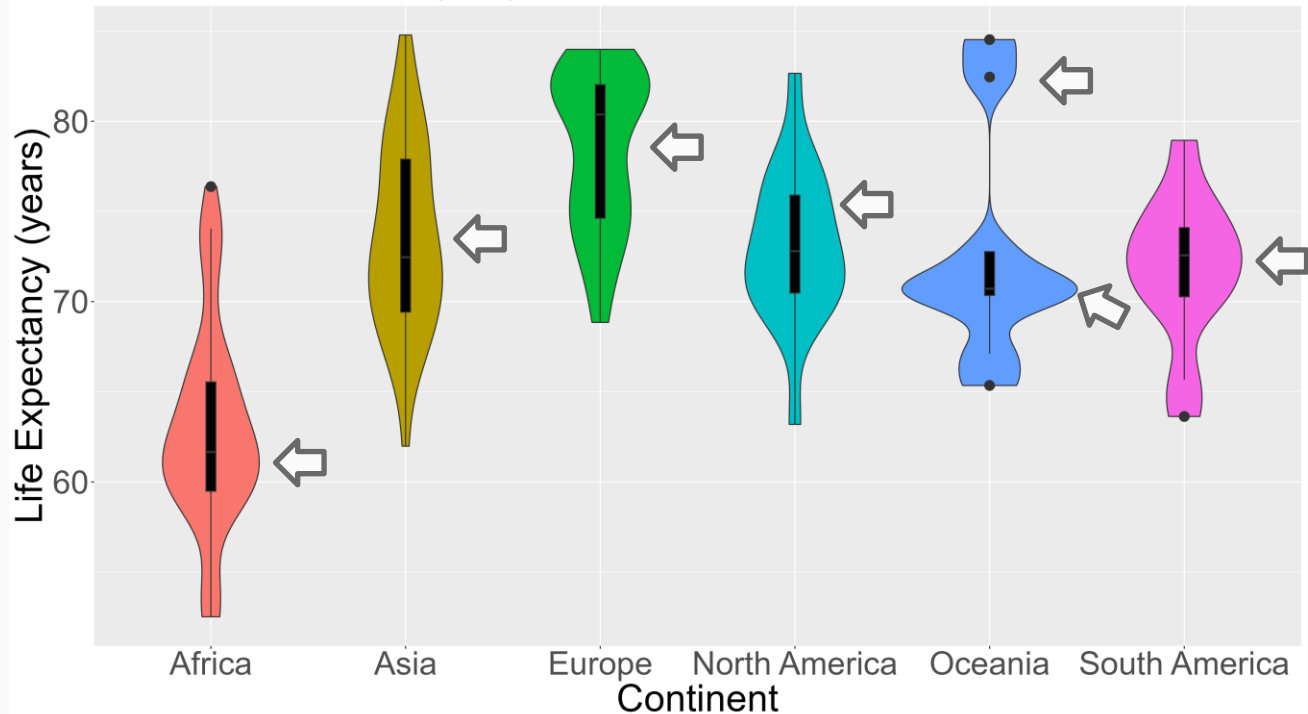


Understand a Violin Plot versus Box Plot



The Shapes of Violin Plots Differ

Life Expectancy by Continent



Categorical Variables

➔ Values that represent category or group membership, like **Favorite Pet** and **Voted**

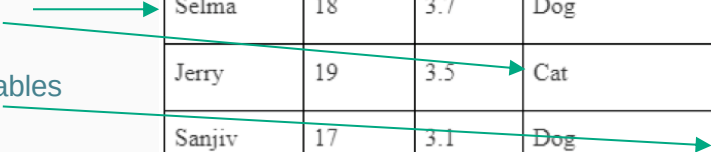
All text variables are categorical variables

Some numeric variables are categorical variables

Mathematical operations are never used on categorical variables



First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0



Categorical Variables

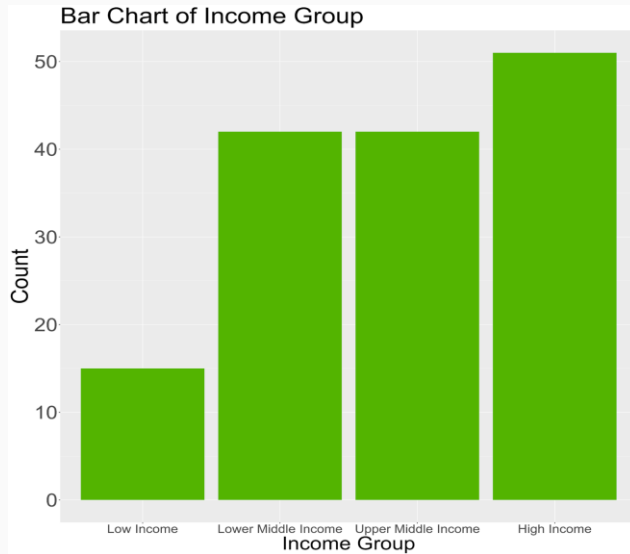
➔ Summarized as **counts** or **percentages**

For example...

- How many people responded “Dog” for Favorite Pet? “Cat”?
- What percentage of people indicated that they voted?

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

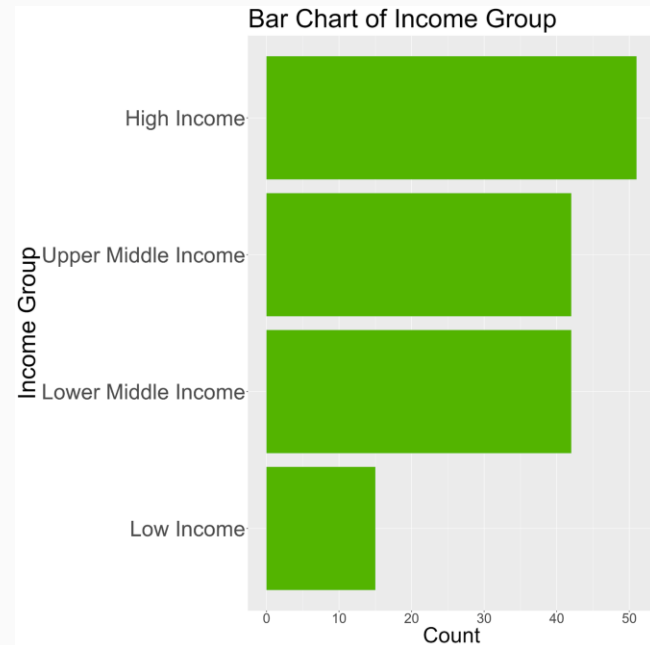
Column Charts



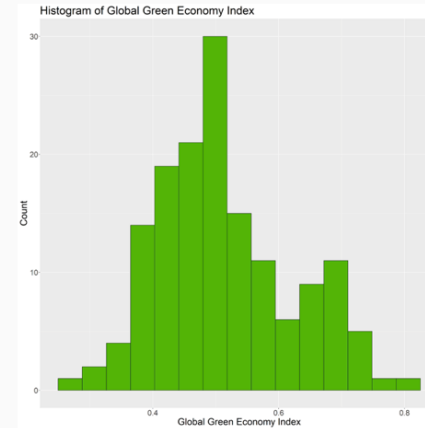
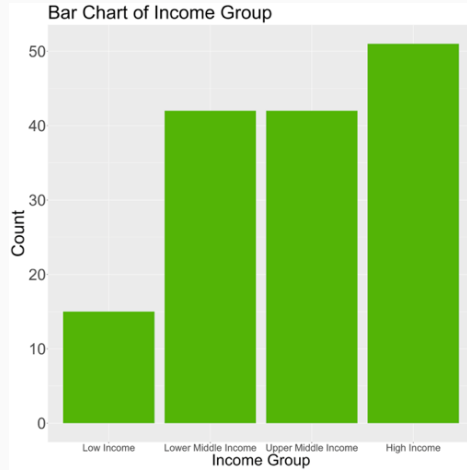
- ✦ Displays a frequency distribution for a categorical variable
- ✦ Shows the number of observations in each category of the variable

Bar Charts

- ✦ Column chart rotated so that the categorical variable is on the y-axis



Bar Charts vs. Histograms



Bar charts and **histograms** both have bars! Remember the differences!

- Column or bar charts for *categorical data*; Histograms for *quantitative data*
- Column or bar charts: space between bars. Histograms: adjacent bins

Word Clouds

- + Collection of words from a set of text data, shown in different sizes
- + Words appearing more frequently in the data appear larger in the cloud

